



Mark Scheme (Results)

January 2022

Pearson BTEC Nationals
In Applied Science (31617H1C)
Unit 1: Principles and Applications of Science I -
Chemistry

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Unit 1: Principles and Applications of Science I

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Marking grids should be applied positively. Learners must be rewarded for what they have shown they can do, rather than be penalised for omissions.
- Examiners should mark according to the marking grid, not according to their perception of where the grade boundaries may lie.
- All marks on the marking grid should be used appropriately.
- All the marks on the marking grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the marking grid.
- Where judgement is required, a marking grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the marking grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The marking grids have been designed to assess learner work holistically. Rows in the grids identify the assessment focus/outcome being targeted. When using a marking grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer, in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band, depending on how they have evidenced each of the descriptor bullet points.

BTEC Next Generation Mark Scheme

Question Number	Answer	Additional Guidance	Mark
1 (a)(i)	soft/conducts electricity/malleable/low density/conducts heat/ductile	allow low boiling point allow shiny when cut	1
1 (a)(ii)	positive charge in circles (1) negative charges outside of the circles (1)	more than one positive and negative charge must be shown mark points are independent reject any negative charges in circles electrons do not need to be in a one-to-one ratio Positive and negative charges do not need to be in one-to-one ratio	2
1 (b)(i)	B – hydrogen		1
1 (b)(ii)	manufacture of {bleach/soap/cleaning materials/paper}/to neutralise acids/drain unblocker/oven cleaner	allow any other correct use	1
1 (b)(iii)	substitution (1) <u>1.50</u> 0.25 evaluation (1) 6 allow other alternative methods	allow 2 marks for 6 with no working	2
1 (c)(i)	$\text{Na (g)} \rightarrow \text{Na}^+ \text{ (g)} + \text{e}^-$ (2) OR (g) (1) e^- (1)	allow e allow –	2

1 (c)(ii)	Award one mark for an identification point and up to two marks for linked expansion points. Any of the following can be an identification point or an expansion point, depending on how the learner shapes their response: electron (being removed) is closer to the nucleus (in lithium atom) (1) less shells/less shielding (1) stronger attraction (of electron) to nucleus (1) electron is harder to remove (1)	allow electron (being removed) is closer to the protons (in lithium atom)	3
Total			12 marks

Question Number	Answer	Additional Guidance	Mark
2 (a)(i)	A – giant ionic		1
2 (a)(ii)	$2\text{Fe} + 3\text{Br}_2 \rightarrow 2\text{FeBr}_3$		1
2 (b)	Identification (electronegativity) increases from gallium to bromine (1) Expansion there is a greater number of protons in the nucleus (1) but the same amount of shielding/same number of inner shells (1) so more attraction for (the bonding pair of) electrons (1)	allow ORA throughout	4
Total			6 marks

Question Number	Answer	Additional Guidance	Mark
3 (a)(i)	C – 1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ¹		1
3 (a)(ii)	loses (1) {one/an} electron (1) accept any other valid response		2
3 (b)	number of moles of KCl (1) $\frac{1.492}{74.6}$ (= 0.02) mass of K required (1) 0.02 x 39.1 Evaluation (1) 0.782	0.782 with no working gains full marks Ecf throughout	3
Total			6 marks

Question number	Indicative content
4	<u>Iron nail</u> - displacement reaction $\text{Fe(s)} + \text{CuSO}_4(\text{aq}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu(s)}$ $\text{Fe} + \text{Cu}^{2+} \rightarrow \text{Fe}^{2+} + \text{Cu}$ - iron is more reactive than copper - iron acts as a reducing agent - therefore displaces copper - copper forms on the nail/is the brown coating - copper ions are the oxidising agent - copper ions remove electrons from iron $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ - copper ions gain electrons and are reduced - iron loses electrons to form iron ions - oxidation state increases therefore is oxidised $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$ <u>Copper coin</u> - copper metal in the coin has the same reactivity as the copper ions and so the copper is not displaced - therefore no oxidation or reduction occurs - copper sulfate solution is blue

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates adequate knowledge of scientific facts/concepts with generalised comments made. • Generic statements may be presented rather than linkages being made so that lines of reasoning are unsupported or partially supported. • The explanation shows some structure and coherence.
Level 2	3–4	• Demonstrates good knowledge and understanding by selecting and applying some relevant scientific knowledge facts/concepts to provide the discussion being presented. • Lines of argument mostly supported through the application of relevant evidence. • The explanation shows a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of scientific facts/concepts to provide the discussion being presented. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence. • The explanation shows a well-developed structure which is clear, coherent and logical.



Llywodraeth Cynulliad Cymru
Welsh Assembly Government

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